

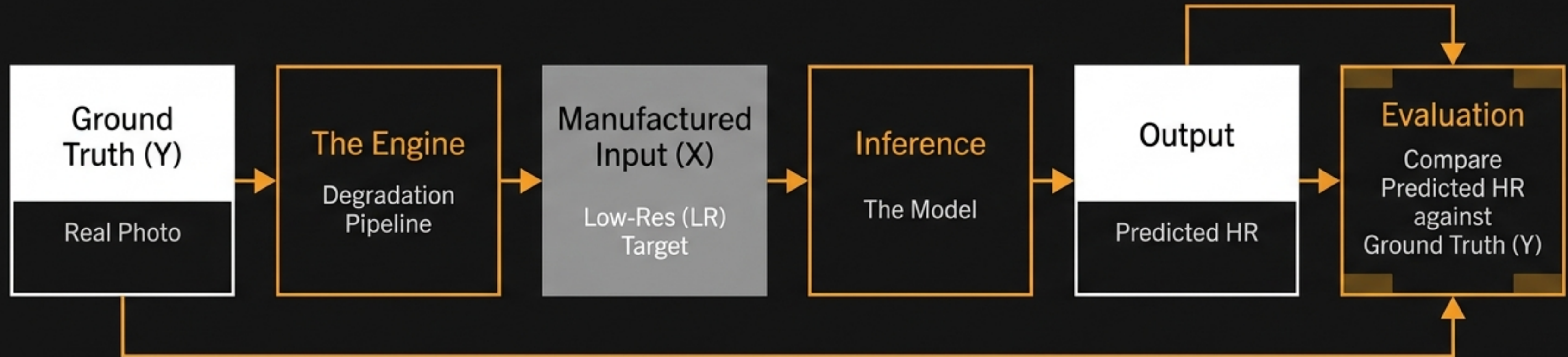
Self-Supervised Image Super-Resolution

Building a 4x upscaler from scratch with no premade dataset.

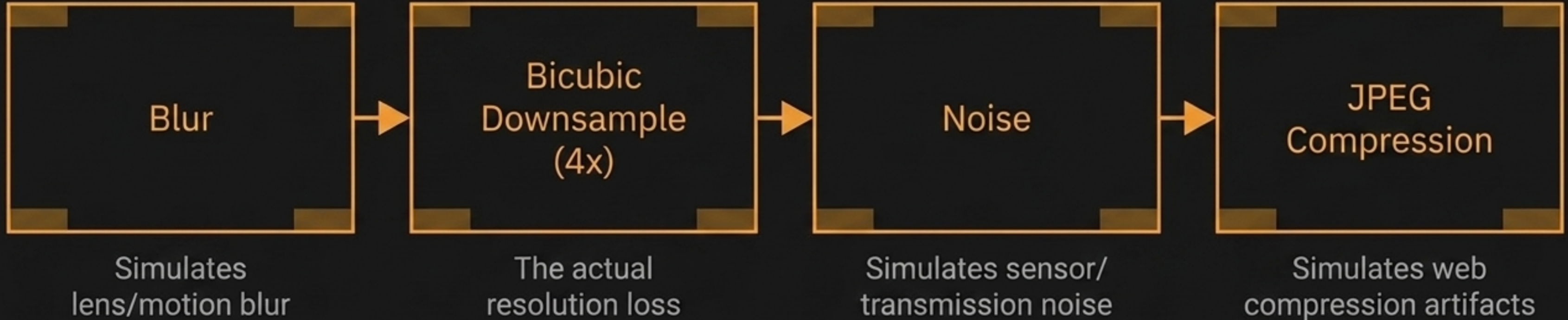


The Trick: Manufacturing the Missing Input

Predict a sharp high-res image from a low-res input—but we only have ordinary photos.



The Degradation Pipeline



Why Randomize?

Parameters are randomized per sample to create a distribution of related problems, forcing the model to learn general detail recovery rather than memorizing one specific blur.

The Risk

Using clean, fixed degradation (like bicubic-only) causes overfitting. The model will fail on real photos with noise or compression.

Baseline Floors to Custom Architecture

Baselines

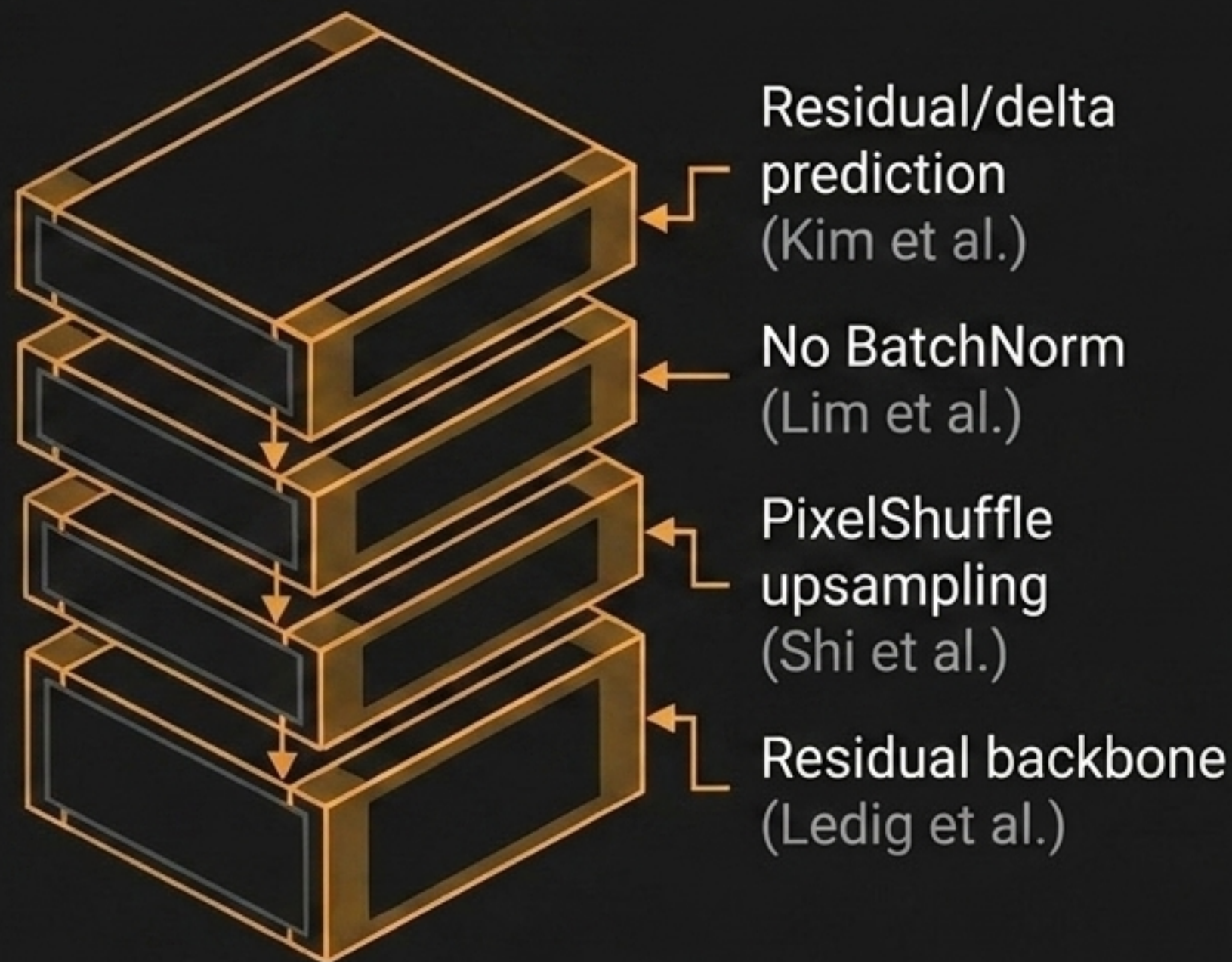
Bicubic

Plain interpolation
(the non-learned floor).

SRCNN

3-layer CNN, direct pixel
prediction (Dong et al.).

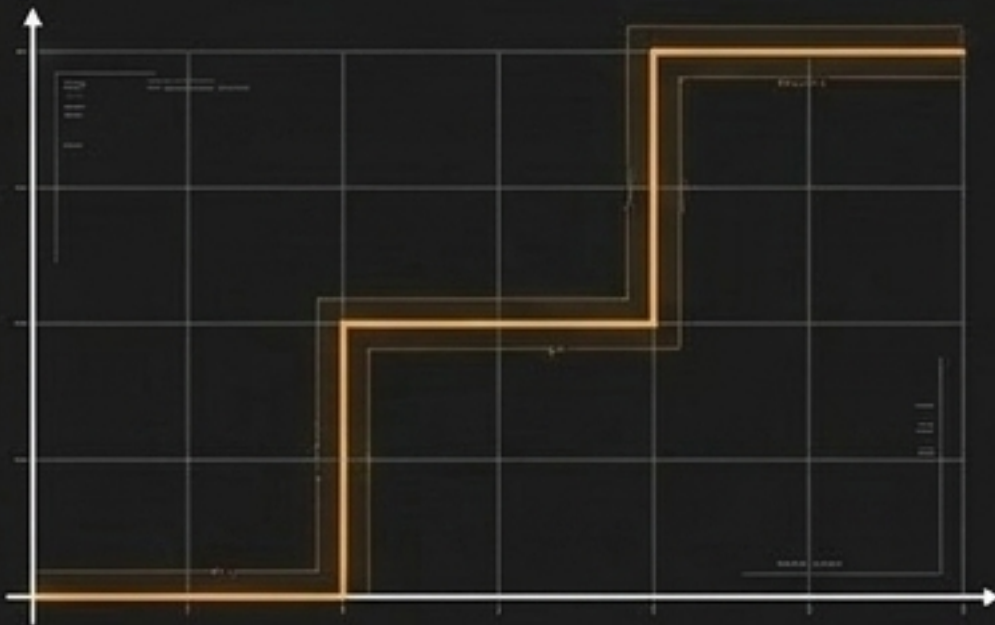
SRResNet-lite Recipe



Optimizing the Objective

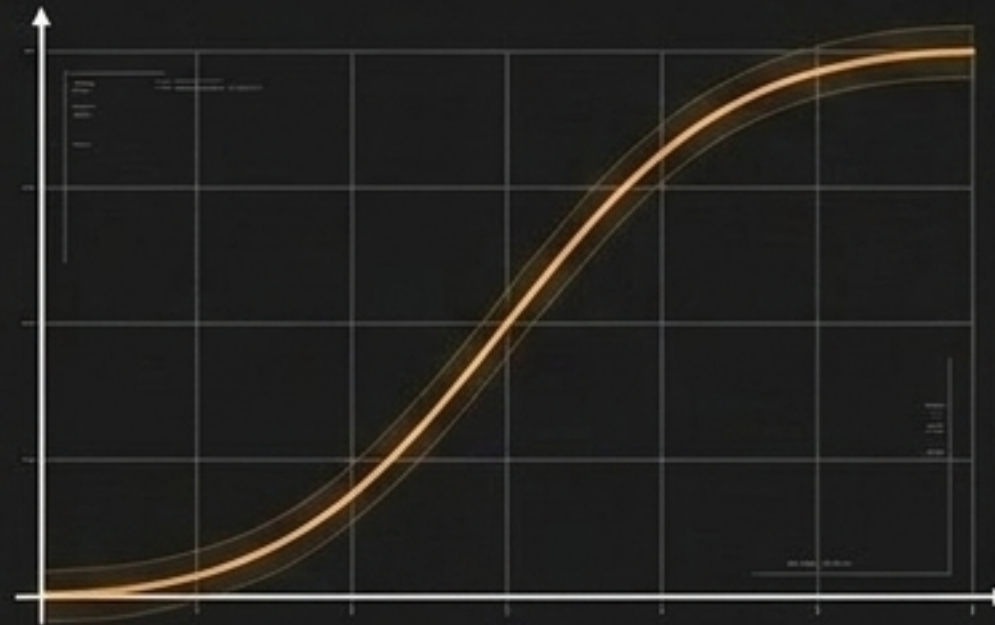
Chosen: L1 Loss

Tracks absolute differences.
Rewards distinct edges and accurate structure.



Rejected: MSE (L2) Loss

Quadratic penalty creates a bias toward a blurry safe average prediction.



Rejected: Adversarial/Perceptual (SRGAN)

Harder to train stably.

Produces visually sharper results, but scores worse on target evaluation metrics (PSNR/SSIM).



The Core Ablation Experiment

Model	PSNR	SSIM
Bicubic (floor)	24.36	0.6366
SRCNN	24.51	0.6452
SRResNet-lite (Bicubic-only)	24.63	0.6431
SRResNet-lite (Full degradation)	25.05	0.6634



The full-degradation model strictly outperforms the bicubic-only baseline, proving realistic degradation is mathematically superior.

Real-World Generalization

The Ultimate Test: Evaluated on genuinely low-quality images lacking a ground truth.

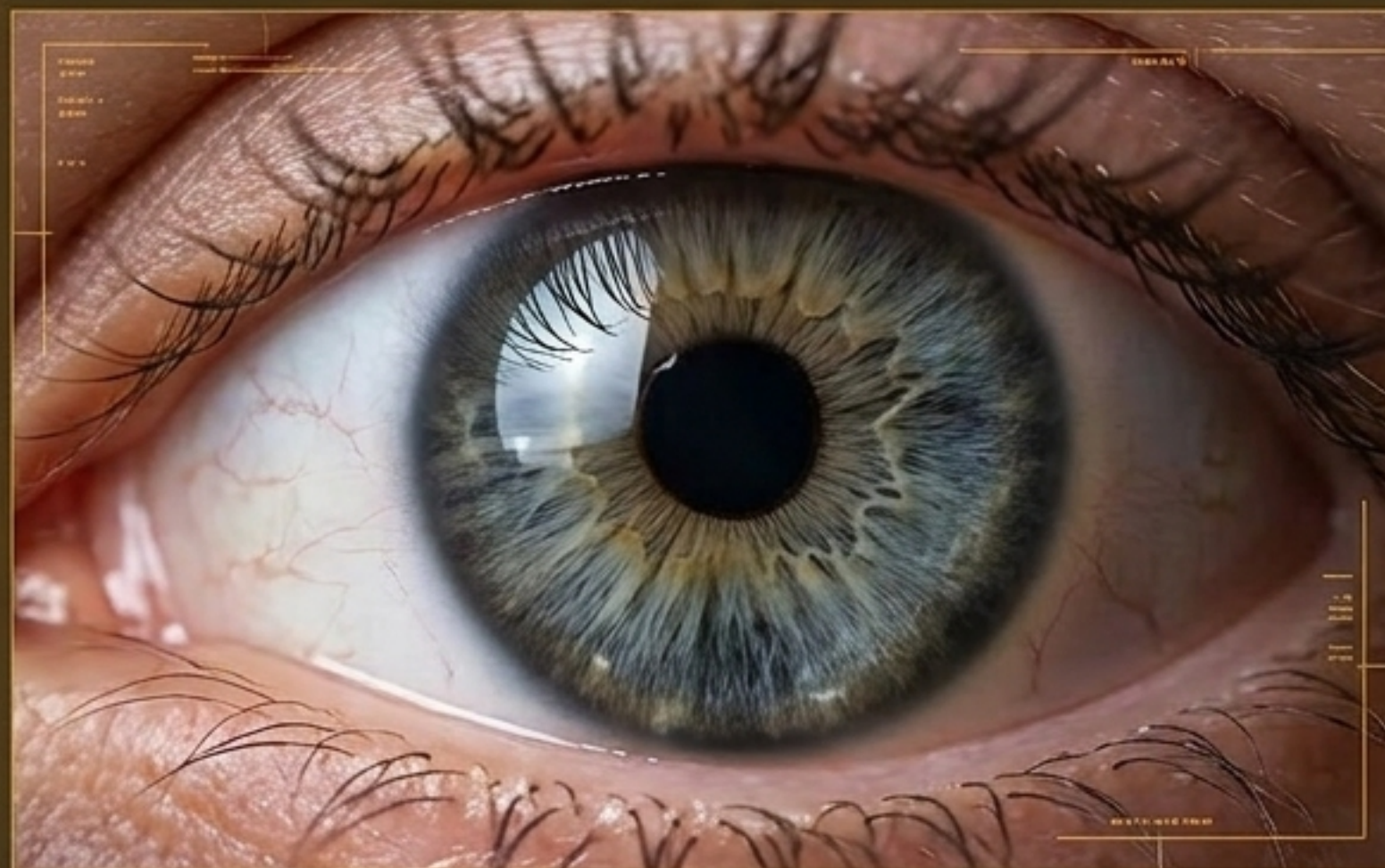
Zero Leakage: These images were never touched by the synthetic training pipeline.

Result: Models trained on clean, bicubic-only data fail on real noise and compression artifacts. The full degradation model thrives.

Clean Data Training (Bicubic-Only)



Full Degradation Training



Future Work & Scaling

Adversarial / Perceptual Loss

Integrating full SRGAN-style training for hyper-realistic textures.

Attention Mechanisms

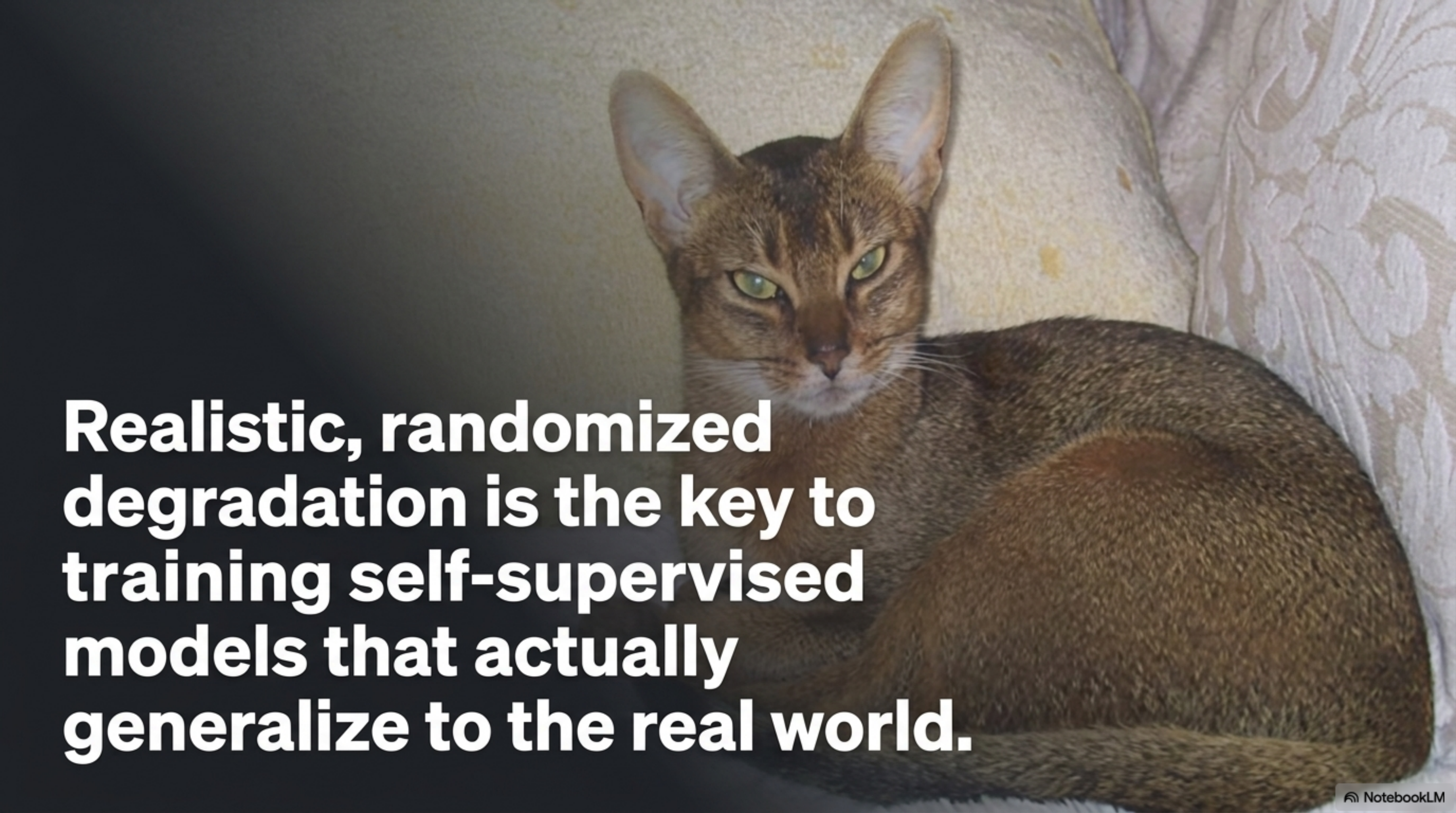
Modernizing the backbone for better spatial awareness.

Blind Degradation Estimation

Predicting degradation parameters dynamically rather than assuming a fixed range.

Expansion

Multi-scale training (8x) and Video Super-Resolution.

A close-up photograph of a brown tabby cat with green eyes, lying down on a light-colored, patterned fabric. The cat is looking directly at the camera with a calm expression. The background is slightly out of focus, showing more of the patterned fabric.

Realistic, randomized degradation is the key to training self-supervised models that actually generalize to the real world.